ECOM155 - Asset Pricing, Trading, and Portfolio Construction - 2023/24

PORTFOLIO ASSIGNMENT

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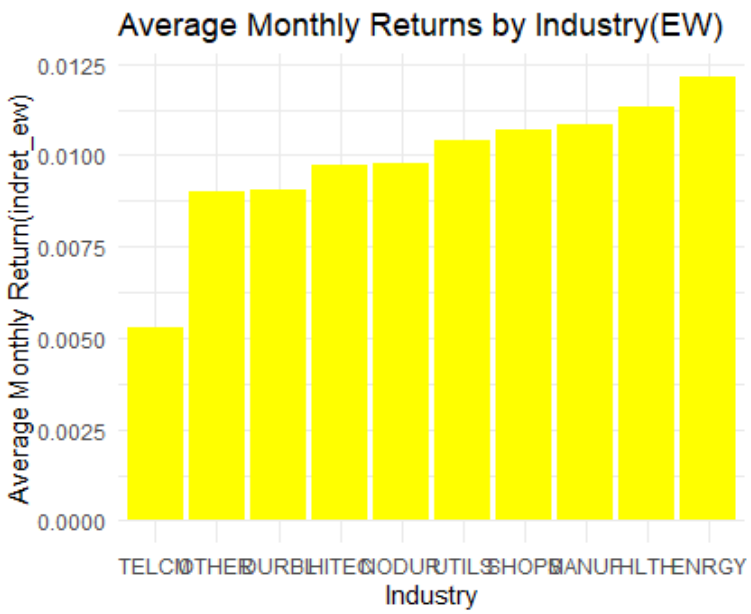
Part A

1.

a.) Based on the average monthly returns using "indret ew" (equal-weighted returns) over the sample period, the results for each industry are as follows:

Industry	**Avg Monthly Return (indret_ew)**
ENRGY	0.01217
HLTH	0.01134
MANUF	0.01086
SHOPS	0.01069
UTILS	0.01040
NODUR	0.00980
HITEC	0.00974
DURBL	0.00904
OTHER	0.00900
TELCM	0.00529



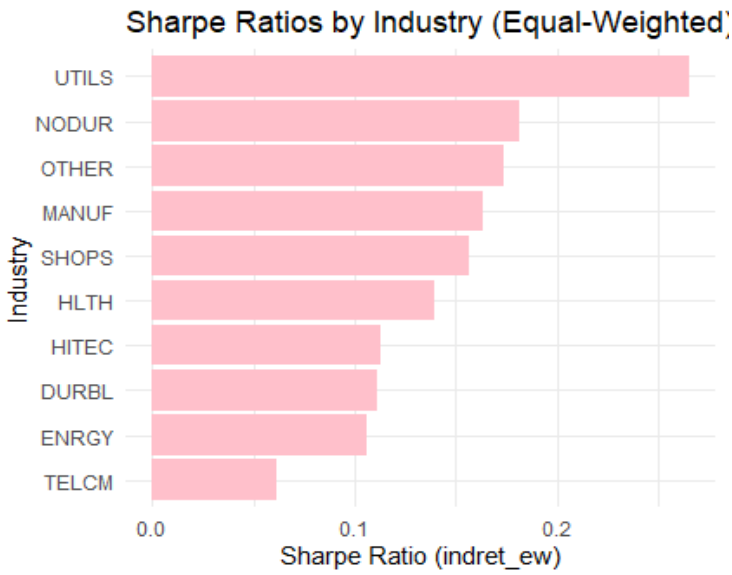
The bar chart visualizes the average monthly returns by industry, using equal-weighted returns (indret_ew). The broad patterns are noticeable:

1**. Rising Trend**: There seems to be a general ascending trend from left to right, with the Telecommunications (TELCM) industry showing the lowest average returns and the Energy (ENRGY) industry showing the highest.

2. **Cluster of Top Performers**: The right side of the chart, representing industries such as Non-Durable (NODUR), Manufacturing (MANUF), Shops (SHOPS), Health (HLTH), and Energy (ENRGY), indicates a cluster of industries with relatively higher returns, suggesting these sectors may offer more robust return profiles over the sample period.

- These patterns suggest that sector-specific factors might be influencing the performance, with certain industries consistently outperforming others in terms of average monthly returns.

b.)



The bar chart presents the Sharpe Ratios for various industries based on equal-weighted returns ("indret_ew"). The Sharpe Ratio is a measure of risk-adjusted return, indicating the excess return earned per unit of volatility or total risk.

In terms of broad patterns across industries:

1**. Differential Risk-Adjusted Returns:** The variation in the length of bars suggests that there is a significant difference in the risk-adjusted returns across industries, with some industries achieving better risk adjusted performance than others.

2. **Utilities Leading**: The Utilities (UTILS) sector stands out with the highest Sharpe Ratio, suggesting that it has provided the most favourable balance of return to volatility among the industries displayed.

3. **Telecommunications at the End**: On the opposite end, the Telecommunications (TELCM) sector shows the lowest Sharpe Ratio, indicating that it has offered less compensation for its volatility compared to other industries.

- These patterns can inform investors about which industries have historically provided better returns per unit of total risk, which is a key consideration in portfolio construction and investment strategy.

c.)

The calculation of the Variance-Covariance matrices for both "indret_ew" and "indret_vw" across the ten sectors has yielded results. Key takeaways include:

1**. Elevated Variance in Equal-Weighted Metrics**: Observing the matrix's diagonal, it becomes apparent that "indret_ew" displays a heightened degree of variance relative to "indret_vw". This points to a greater dispersion of returns in portfolios where each stock is weighted equally. Often, portfolios weighted by value will be more heavily influenced by larger firms, which typically exhibit less price fluctuation.

2. **Patterns in Covariance**: Inter-industry covariance is more pronounced in the equal-weighted returns, as seen in the off-diagonal elements. This might reflect a synchronized movement amongst industries within equal-weighted portfolios, possibly because smaller companies, which have a stronger presence in such portfolios, respond more homogeneously to overarching market events.

3. **Divergence in Matrix Relationships**: A discernible difference emerges when comparing the two matrices; the matrix for "indret_ew" reflects more significant inter-industry connections than that of "indret_vw". The latter may be shaped to a greater extent by larger corporations, whose returns are affected by distinct factors not as prevalent among smaller entities.

The contrast between the matrices is likely rooted in the distinct approaches to weighting. Equal-weighted approaches distribute importance uniformly across all stocks in an industry, which can highlight sector-specific risk factors. Conversely, value-weighted approaches tend to emphasize the influence of larger firms, which may alter the risk and return dynamic within the industry spectrum. Thus, variances and covariances in value-weighted returns may be more representative of the market behaviour of these larger entities.

d.)

Based on the bar plots of the market betas of the industries using both "indret ew" (equal-weighted returns) and "indret vw" (value-weighted returns), we identify the highest and lowest industry market beta.

<u>For "indret ew" (equal-weighted returns), as seen in the blue bar plot:</u>

- The Energy (ENRGY) industry has the highest market beta.
- The Utilities (UTILS) industry has the lowest market beta.

<u>For "indret vw" (value-weighted returns), as seen in the green bar plot:</u>

- The Durable (DURBL) industry has the highest market beta.
- Again, the Utilities (UTILS) industry has the lowest market beta.

A comparative look at the two graphical representations reveals that the Utilities sector consistently exhibits the least market beta under both weighting methods, yet the sector with the most market beta diverges: Energy leads under equal-weighted returns, whereas Durables lead under value-weighted returns. This variance underscores how different weighting approaches can shift the perceived market risk across sectors.

The heightened beta of the Energy sector for equal-weighted returns might suggest that smaller firms in this bracket experience greater market volatility. On the flip side, the pronounced beta for Durables under value-weighted returns may indicate that the larger firms, which have a greater impact on the value-weighted index, also experience greater volatility when gauged against the market. These differences highlight how the size and

weighting of companies within an industry can affect the industry's overall risk profile as measured by market beta.

2. Predictive Regressions.

a.)

i. **Significance of Predictive Indicators**: Indicators that have a noteworthy level of predictive capability are identified by their p-values being lower than the predetermined threshold for statistical significance.

ii. **Widespread Predictors Across Sectors**: Predictive indicators that are consistently significant in regression analyses across various industry segments can be considered widespread predictors.

iii. **Deciphering Coefficients**: The polarity of a coefficient (be it positive or negative) points to the nature of the correlation with forthcoming industry yields. Specifically, a positive coefficient implies an increase in the financial indicator is linked with an uptick in expected industry returns for the succeeding period, and the reverse holds for a negative coefficient.

- **Positive Coefficient Implications**: A higher metric can hint at a rosier outlook for future performance. For instance, a robust current ratio may signal strong liquidity, potentially foreshadowing more favourable future yields.
- **Implications of Negative Coefficients**: In contrast, a higher metric here could be a harbinger of less favourable future performance. For example, a substantial debt-to-equity metric might be indicative of elevated risk, which could be perceived unfavourably by market participants.

Scenario with No Significant Indicators: A scenario where financial indicators fail to predict future returns might occur for several reasons. It could suggest market efficiency, where all accessible information is already reflected in current prices, rendering these indicators non-predictive. Alternatively, it could also mean that the financial ratios fail to encapsulate the drivers of future returns, or that the time frame under review does not provide a comprehensive reflection of typical conditions.

b.)

The predictors vary across industries, reflecting the different financial drivers that might influence returns in different market segments. The best predictors for each industry, determined by the lowest Root Mean Squared Error (RMSE), are as follows:

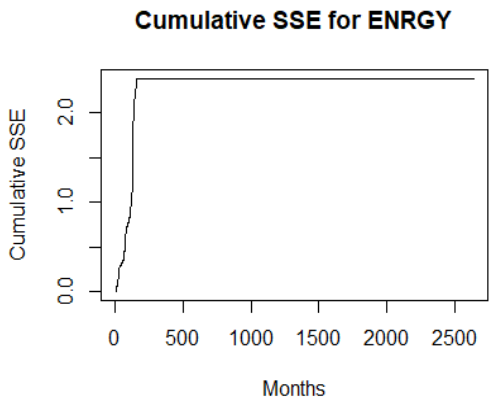
- **Durable Goods (DURBL)**: Book-to-Market Ratio (RMSE: 0.0804)
- **Energy (ENRGY)**: Short-Term Debt (RMSE: 0.1132)
- **High Technology (HITEC)**: Staff Sale Ratio (RMSE: 0.0886)
- **Health (HLTH)**: Book-to-Market Ratio (RMSE: 0.0801)
- **Manufacturing (MANUF)**: Dividend Yield (RMSE: 0.0675)
- **Non-Durable Goods (NODUR)**: Gross Profit Margin (RMSE: 0.0541)
- **Other (OTHER)**: Interest-Bearing Debt (RMSE: 0.0522)
- **Shops (SHOPS)**: Book-to-Market Ratio (RMSE: 0.0695)
- **Telecommunications (TELCM)**: Debt to EBITDA (RMSE: 0.0878)
- **Utilities (UTILS)**: Cash to Long-Term Debt (RMSE: 0.0365)

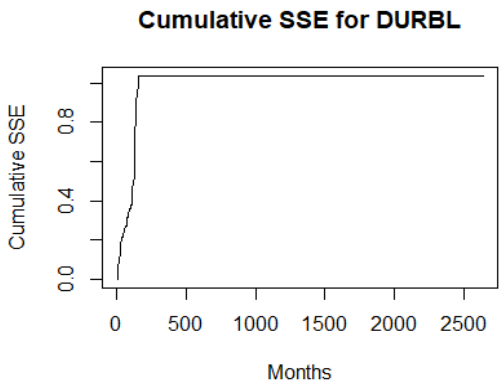
These predictors represent the most effective financial ratios for forecasting next period's returns in their respective industries.

PART B

3.)

a.)


Cumulative SSE for ENRGY
Cumulative SSE
0.0
1.0
2.0
0
500
1000
1500
2000
2500
Months


Cumulative SSE for DURBL
Cumulative SSE
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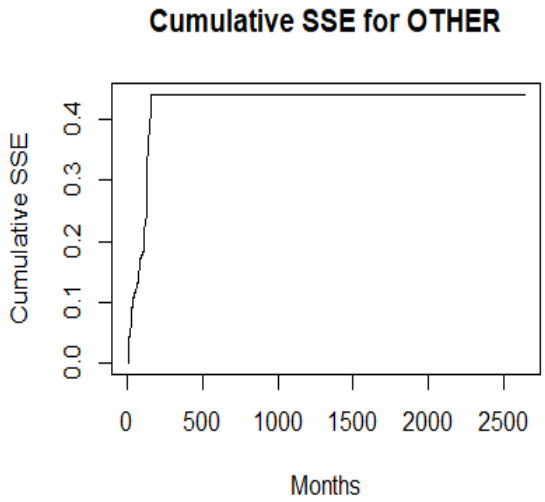
0.4
0.8
0
500
1000
1500
2000
2500
Months


Cumulative SSE for OTHER
Cumulative SSE
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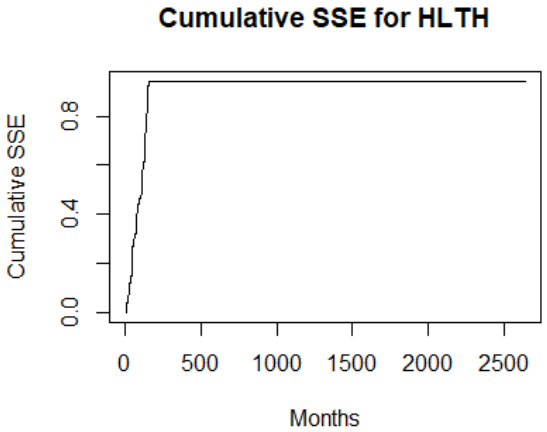
0.3
0.4
0
500
1000
1500
2000
2500
Months


Cumulative SSE for HLTH
Cumulative SSE
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0
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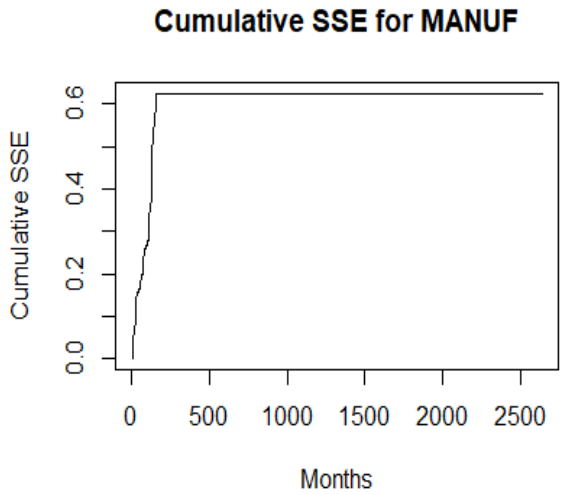
1000
1500
2000
2500
Months


Cumulative SSE for MANUF
Cumulative SSE
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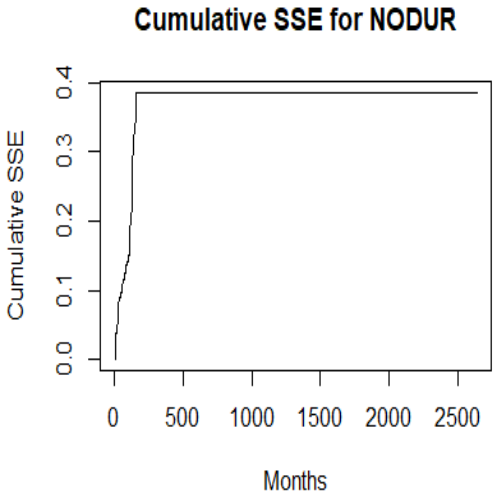
0.6
0
500
1000
1500
2000
2500
Months


Cumulative SSE for NODUR
Cumulative SSE
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0.3
0.4
0
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1000
1500
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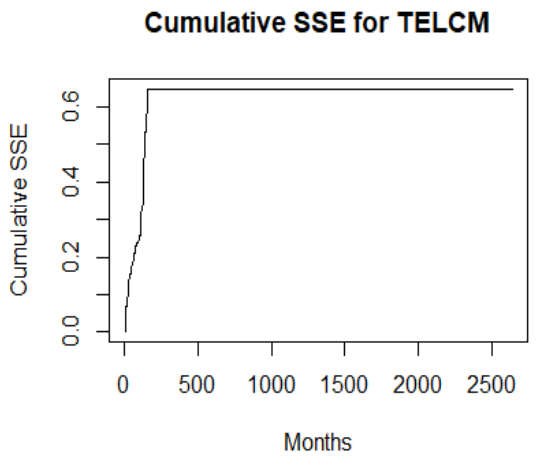
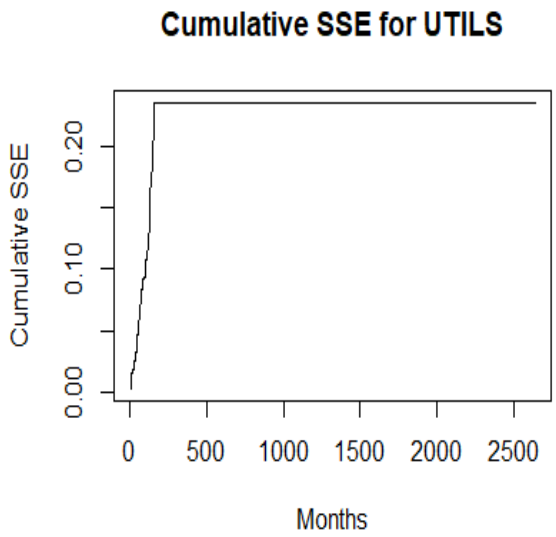
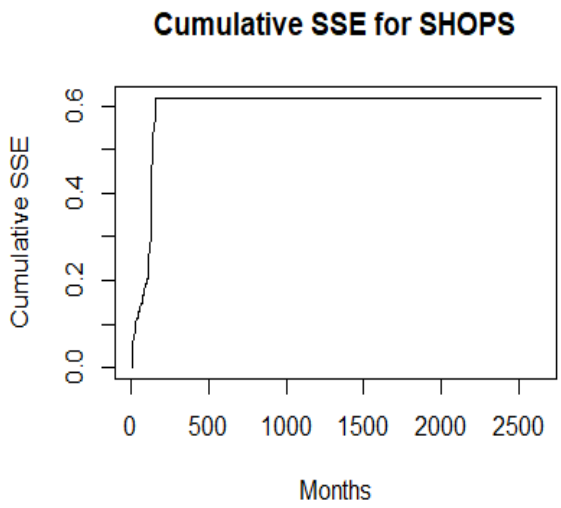
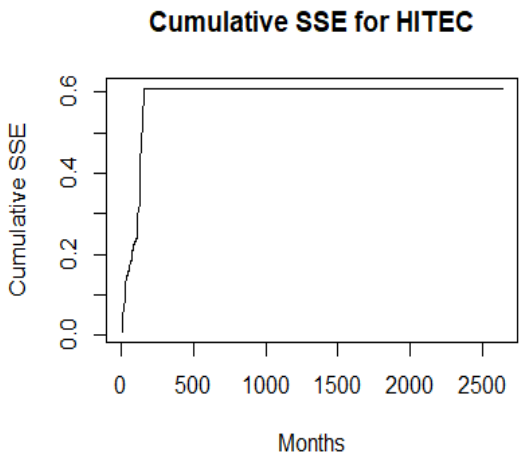
2500
Months









Analysing the cumulative SSE graphs for 10 industries:

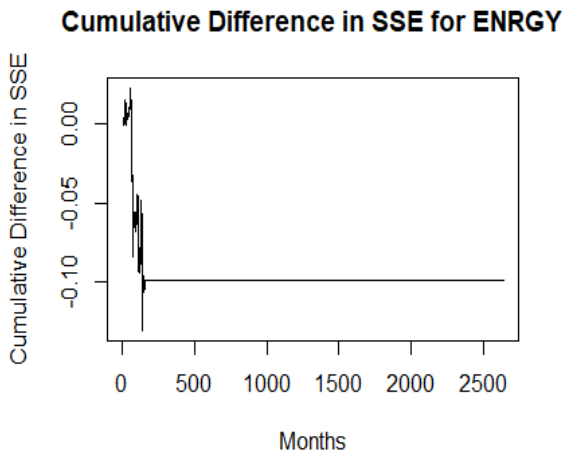
i. **Assessing the Predictor**: An effective predictor is indicated by a lower and stabilizing cumulative SSE. This trend means the forecast errors are minimal and consistent.

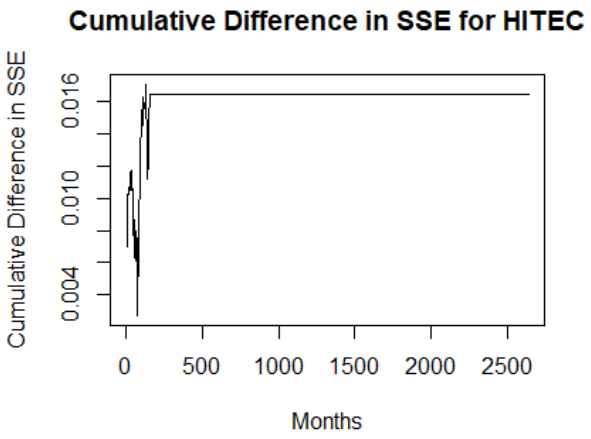
ii. **Investment Decision**: The optimal sector for investment would be one with a minimal cumulative SSE, suggesting more precise forecasts. The curve's plateau towards the later months would imply a solidified model with dependable predictions.

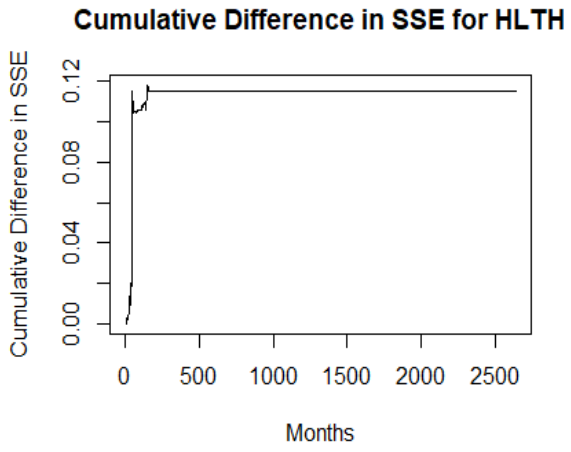
While a low SSE signals precise forecasting, it does not equate to high returns. Investment choices should also weigh return rates, risks, and sector prospects.

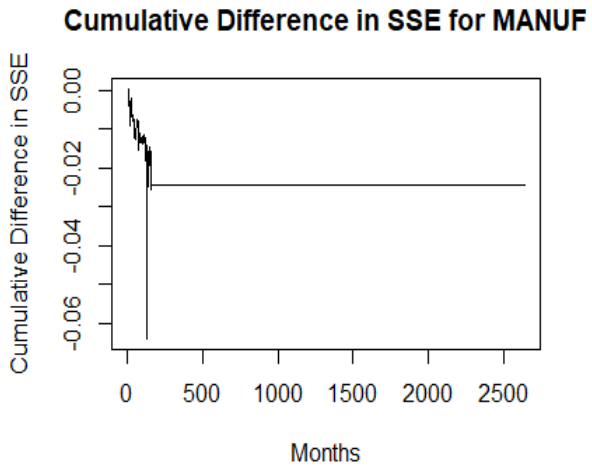
From the visuals, a sector like Utilities, exhibiting a minimal and level SSE trajectory, may denote steady returns, potentially marking it as a solid choice for investment, assuming the returns are substantial.

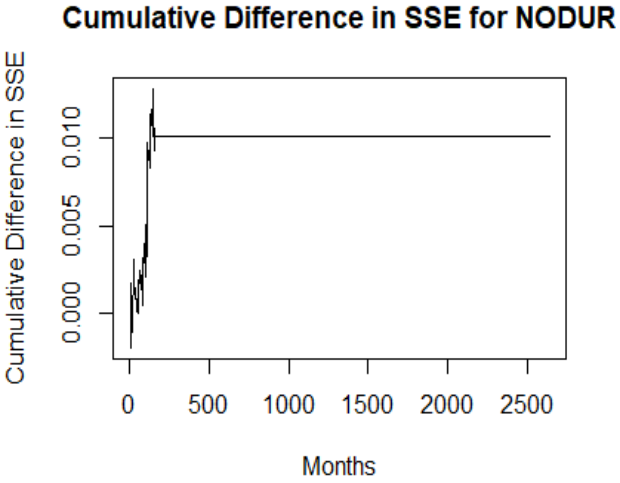
b.)

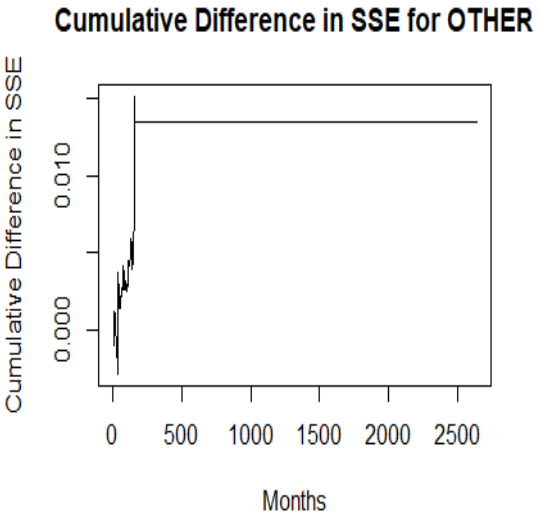


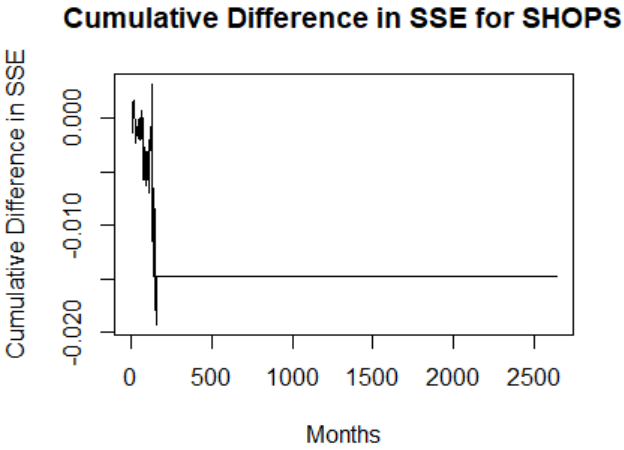


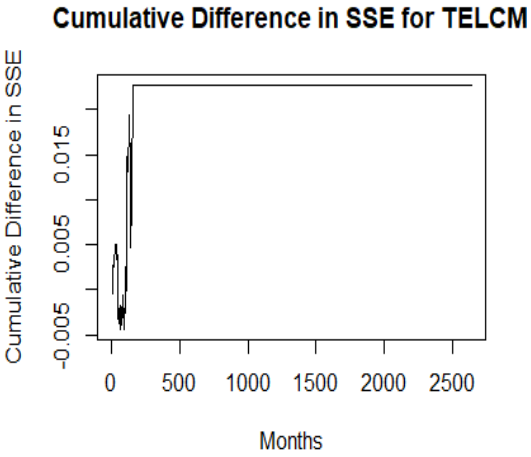


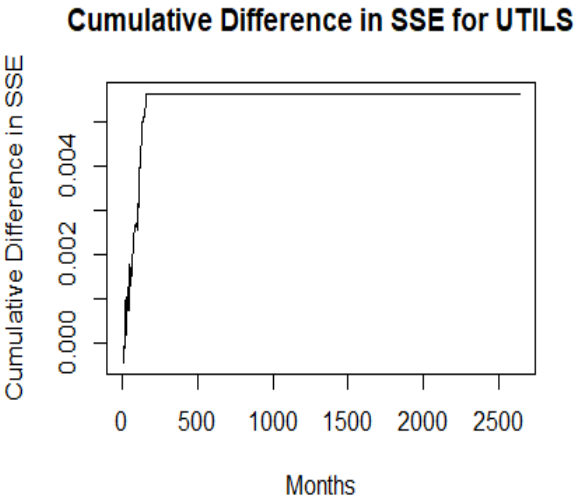


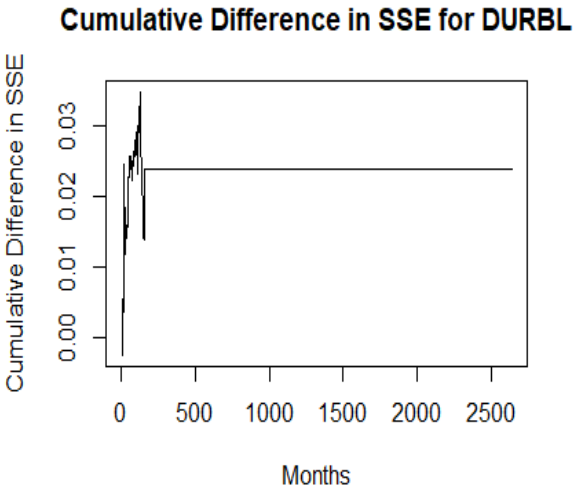












In an efficient market, the mean return of a portfolio is often used as a benchmark because it is assumed that all available information is already factored into the prices,

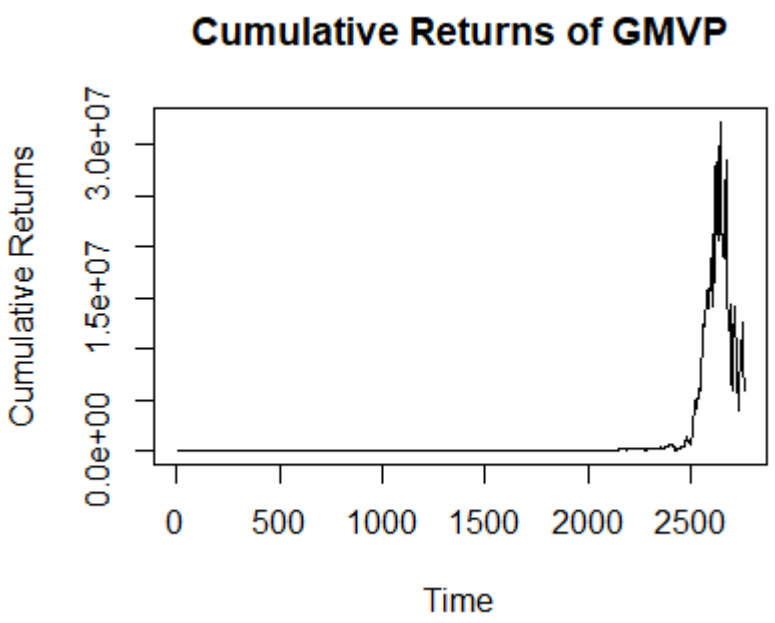
and thus future prices should only react to new information, which by its nature is unpredictable. The mean return serves as a simple prediction model against which more complex models can be compared.

A negative cumulative difference would indicate that the model has a lower SSE compared to the benchmark and is doing a better job of predicting returns.

Conversely, a positive cumulative difference would indicate that the model has a higher SSE than the benchmark, suggesting that it is not adding predictive value beyond what the mean return provides.

4.)

a.)



Cumulative Returns Interpretation:

- The plot shows that the portfolio's value has increased over time, which is positive.
- The slope of the cumulative return curve provides insight into the growth rate; a steeper slope indicates faster growth.
- Any flat periods or sharp declines in the plot represent times of stagnation or losses in the portfolio value.

Annualized Sharpe Ratio:

- An annualized Sharpe Ratio of 1.943 is quite high. Typically, a Sharpe Ratio above 1 is considered good, above 2 is very good, and above 3 is excellent.
- This ratio indicates that the portfolio has provided a high level of excess return per unit of risk.

- Since the Sharpe Ratio is a risk-adjusted measure, this high ratio suggests that the returns of the portfolio have compensated well for the volatility experienced over the period.

THE END